

Requirement Analysis

Customer Journey Map — Priya (Skincare Consumer)

Field	Details
Team Lead	Govindu Kavya Sri
Team Members	Eswari Gayatri Budumuru, Kiladi Kalavathi, Adaka Mounika, Panditi Jyostna
Domain	Retail / Cosmetics Analytics
Tools Used	Python, Microsoft Excel, Tableau Public
Dataset	Cosmetics product dataset — 1,472 records, 11 attributes
Published Dashboard	public.tableau.com/app/profile/govindu.kavya.sri/viz/cosmeticproject_17872016666220/Dashboard1

Project Context

The cosmetics industry is one of the fastest-growing and highly competitive sectors, driven by changing consumer preferences, diverse skin types, brand innovation, and evolving pricing strategies. With increasing demand for personalized beauty products, companies rely on data-driven insights to understand market trends, consumer behavior, and product performance.

This project, “Cosmetic Insights: Navigating Cosmetics Trends and Consumer Behavior with Tableau,” focuses on analyzing a cosmetics dataset to uncover meaningful patterns and actionable insights. The dataset includes key attributes such as brand, product category, price, rank, and skin type suitability (Dry, Oily, Combination, Normal, and Sensitive).

This project is designed to help businesses and analysts understand cosmetics market trends and consumer preferences. By analyzing brand performance, pricing patterns, product distribution, and skin-type coverage, the dashboard enables stakeholders to identify high-performing products, target specific customer segments, and optimize marketing and product strategies.

Scenario 1: Monitoring Consumer Preferences

In a real-time scenario, imagine receiving an alert indicating a concerning trend in consumer preferences, such as a significant decline in interest in certain cosmetic products or ingredients. Using the Cosmetic Insights data, we can promptly assess the extent and potential impact of this trend, identify contributing factors, and deploy immediate interventions to adapt product offerings and marketing strategies. Whether

it's through targeted promotional campaigns, adjustments in product formulations, or personalized recommendations, real-time analysis enables agile decision-making and proactive measures to meet evolving consumer needs.

Scenario 2: Addressing Product Concerns

In the event of identifying widespread product concerns, such as negative reviews or safety issues associated with specific cosmetic items, real-time access to Cosmetic Insights data enables swift response and management. Cosmetic companies and regulatory bodies can utilize the dataset to gather crucial information about the concerns, including their prevalence, potential impacts on consumer trust, and affected product demographics. By leveraging real-time analytics, they can implement quality control measures, recall products if necessary, and communicate transparently with consumers to address their concerns and maintain brand integrity.

Scenario 3: Predictive Analysis and Product Innovation

Leveraging predictive analytics capabilities, Cosmetic Insights empowers companies to anticipate and respond to emerging trends and consumer preferences in the beauty industry. By analyzing historical data and identifying predictive indicators, companies can proactively innovate new products, adjust existing formulations, and tailor marketing strategies to meet evolving consumer demands. Real-time monitoring of market trends, consumer feedback, and competitor activities enables timely interventions, product innovation, and strategic decision-making to stay ahead in a competitive market landscape.

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Stage	Awareness	Consideration	Onboarding	Usage	Decision / Outcome
Actions	Realizes she keeps buying products that don't suit her skin	Hears about a dashboard comparing cosmetics by price, rating, and skin type	Opens the Tableau Public dashboard for the first time via a shared link	Clicks a category bar and watches the scatter plot, brand ranking, and heatmap filter together; narrows the Brand filter	Chooses a product/brand based on visible price-to-rating value and confirmed skin-type suitability
Thoughts	"Why do I keep wasting money on the wrong products?"	"Could a dashboard actually help me compare options?"	"This is clean and simple — no login needed"	"This brand is well-rated and reasonably priced for my skin type"	"I feel confident this is a good purchase"
Emotions	Frustrated, uncertain	Curious, slightly skeptical	Reassured	Engaged, empowered	Confident, satisfied
Touchpoints	Personal shopping experience, social media	Shared dashboard link	Tableau Public dashboard (shared link)	Interactive filters: Category click, Brand search	Final purchase decision informed by data

Stage	Awareness	Consideration	Onboarding	Usage	Decision / Outcome
Opportunities	Raise awareness of the dashboard among skincare communities	Keep the dashboard link simple and login-free	Ensure the title and first chart orient a first-time viewer instantly	Ensure cross-chart filtering responds in under 2 seconds	Track which categories/brands are explored most, to guide future updates

The emotional curve dips lowest at Awareness/Consideration, then climbs steadily once Priya reaches Onboarding and Usage — confirming that a clean, self-explanatory first chart and responsive cross-chart filtering are the difference between a visitor abandoning the dashboard and one who reaches a confident purchase decision. Two specific requirements below were added directly because of this journey: cross-chart filter actions (FR-09) and a searchable Brand filter (FR-10), both tracing back to friction points identified at the Usage and Decision stages.